

Assignment Module 8:

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Network Access Basic routing and advance routing concept, switching concept-

1. Explain Switch

ANS: - A switch is a network device that connects multiple computers in LAN and forwards data only to the correct device using MAC addresses.

2. Explain Switch Boot Sequence

ANS: - Switch Boot Sequence:

1. POST (Power on Self-Test)
2. Load Boot Loader
3. Load IOS (from Flash)
4. Load Startup-config (from NVRAM)

3. Explain Three Methods to Access Switch CLI

ANS: - Three methods to access Switch CLI:

1. Console Access – Using console cable
2. Telnet – Remote access (unencrypted)
3. SSH – Remote secure access

4. Explain and Configure Cisco IOS

ANS: - Cisco IOS is the operating system used in Cisco network devices. It controls switching, routing, security.

Basic configuration:

- Switch> enable
 - Switch# configure terminal
 - Switch(config)# hostname HR
 - Switch(config)# enable secret SUMIT
 - Switch(config) line console 0
 - Switch(config-line) password SUMIT
 - Switch(config-line) login
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5. Explain Switch Port

ANS: - A switch port is a physical interface used to connect devices like PCs, routers, access points. Each port works as an individual collision domain.

6. R1, R2, R3, and R4 have their Fast Ethernet 0/0 interfaces attached to the same VLAN. A network engineer has typed a configuration for each router by using a word processor. He will later copy and paste the configuration into the routers. Examine the following exhibit, which lists configuration for the four routers, as typed by the network engineer. Assuming that all four routers can ping each other's LAN IP addresses after the configuration has been applied, choose the routers that will be able to form a neighbor relationship with the other routers on the LAN. (You can assume that, if not shown in the exhibit, all other related parameters are still set to their defaults.) (Choose two)

- A. R1
- B. R2
- C. R3
- D. R4
- E. None of the routers will exchange routing information

ANS: - (A)R1

(C) R3

- R1 and R3 have matching OSPF settings (same area, same timers, no configuration conflicts), so they can form a neighbor relationship.
 - R2 and R4 have OSPF mismatches (like different area/timers/authentication/router-ID issues), so they will **not** form adjacency.
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7. enable secret [password] is hashed using the algorithm.

A. MD5

B. AH

C. PSK

D. ESP

E. WPA2

ANS: - (A) MD5

- Cisco enable secret password is encrypted using **MD5 hashing**.
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8. An engineer connects to Router R1 and issues a show ip ospf neighbor command. The status of neighbor 2.2.2.2 lists FULL/BDR. What does the BDR mean?

A. R1 is an Area Border Router.

B. R1 is a backup designated router.

C. Router 2.2.2.2 is an Area Border Router.

D. Router 2.2.2.2 is a backup designated router.

ANS: - (D) Router 2.2.2.2 is a backup designated router.

- BDR = Backup Designated Router in OSPF.
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8. Which command is used to view the neighbor discovery table on a PC?

- A. show ipv6 neighbor
- B. show ipv6 neighbors
- C. netsh interface ipv6 show neighbor
- D. netsh interface ipv6 show neighbors

ANS :- (C) netsh interface ipv6 show neighbor

- Windows uses **netsh** command for IPv6 neighbor table.
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9. What type of variable is being shown? Routers = [R1,R2,R3]

- A. List
- B. Dictionary
- C. Simple
- D. Unsigned integers

ANS: - (A) List

10. Identify the fields in an IPv4 header. (Choose three)

- A. Host component
- B. Time to Live
- C. Source address
- D. Destination address

ANS: - (B) Time to Live

(C) Source address

(D) Destination address

- These are real IPv4 header fields
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